Gossip Simulator

Measured convergence of asynchronous Erlang actors

Team and scope

Omkar Salkade and Paresh Devlekar. Both gossip and push-sum run on full, 2D grid, line and imperfect 2D grid networks. Each vertex is a separate Erlang process; communication and coordination use asynchronous messages only.

What counts as convergence?

Gossip: every actor has received the rumor at least once. Each informed actor sends to a random neighbor on a 1 ms timer and stops transmitting after ten receptions. Coverage convergence may precede local transmission termination. It is possible for a rumor to die out before full coverage; such a run is reported as a timeout.

Push-sum: the first actor to complete three consecutive receptions with an absolute ratio change below 1e-10 terminates, and its notification ends the measured run. This is first local convergence, not proof of network-wide convergence. Initially s=i and w=1. Every receive adds the incoming mass, retains half, and sends half to a random neighbor. Only one actor is initially started. There is no periodic push-sum sender.

The assignment specifies local termination but leaves global convergence unspecified. The default uses literal actor exit and first local termination: a single receive-driven message chain can otherwise disappear at its first terminated actor. An optional global mode preserves mass by keeping stable actors forwarding until all are stable; it is not used for these measurements.

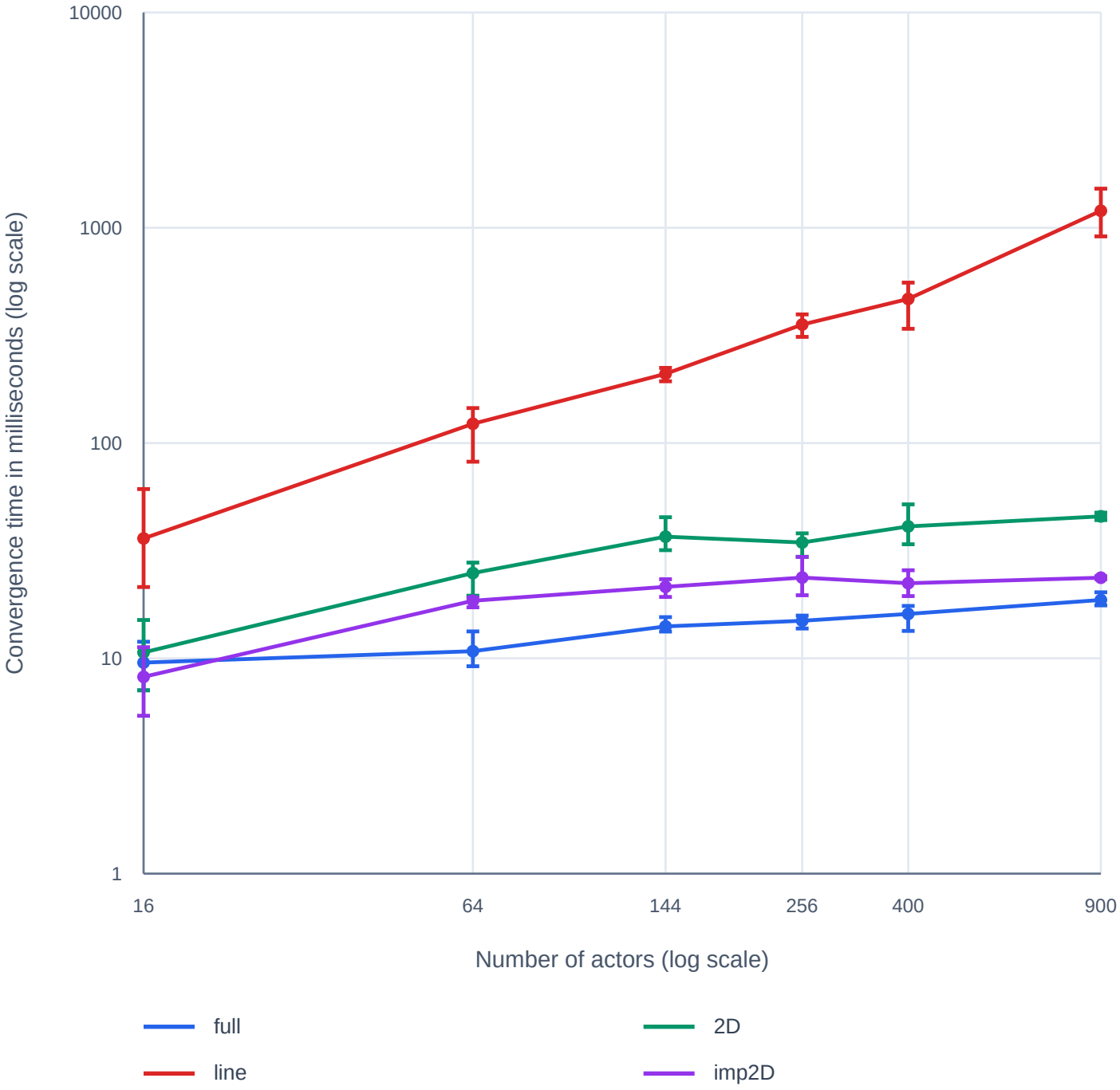
Measurement protocol

144 serial runs: 3 seeds per size, topology and algorithm; sizes 16, 64, 144, 256, 400, 900. Timeout: 8 s per run. Seeds are 1000 + trial. Timing uses erlang:monotonic_time(microsecond), from the first start message through the selected convergence notification. VM launch, compilation, actor/topology setup and cleanup are excluded.

Host: Darwin-25.0.0-arm64-arm-64bit-Mach-O; arm64; Erlang/OTP 29; four VM schedulers. A seed fixes random streams but does not fix asynchronous scheduling. Curves show arithmetic means only where every trial succeeded; vertical bars show min/max. Failed sizes are marked at the timeout threshold, not treated as measured convergence.

Rumor propagation

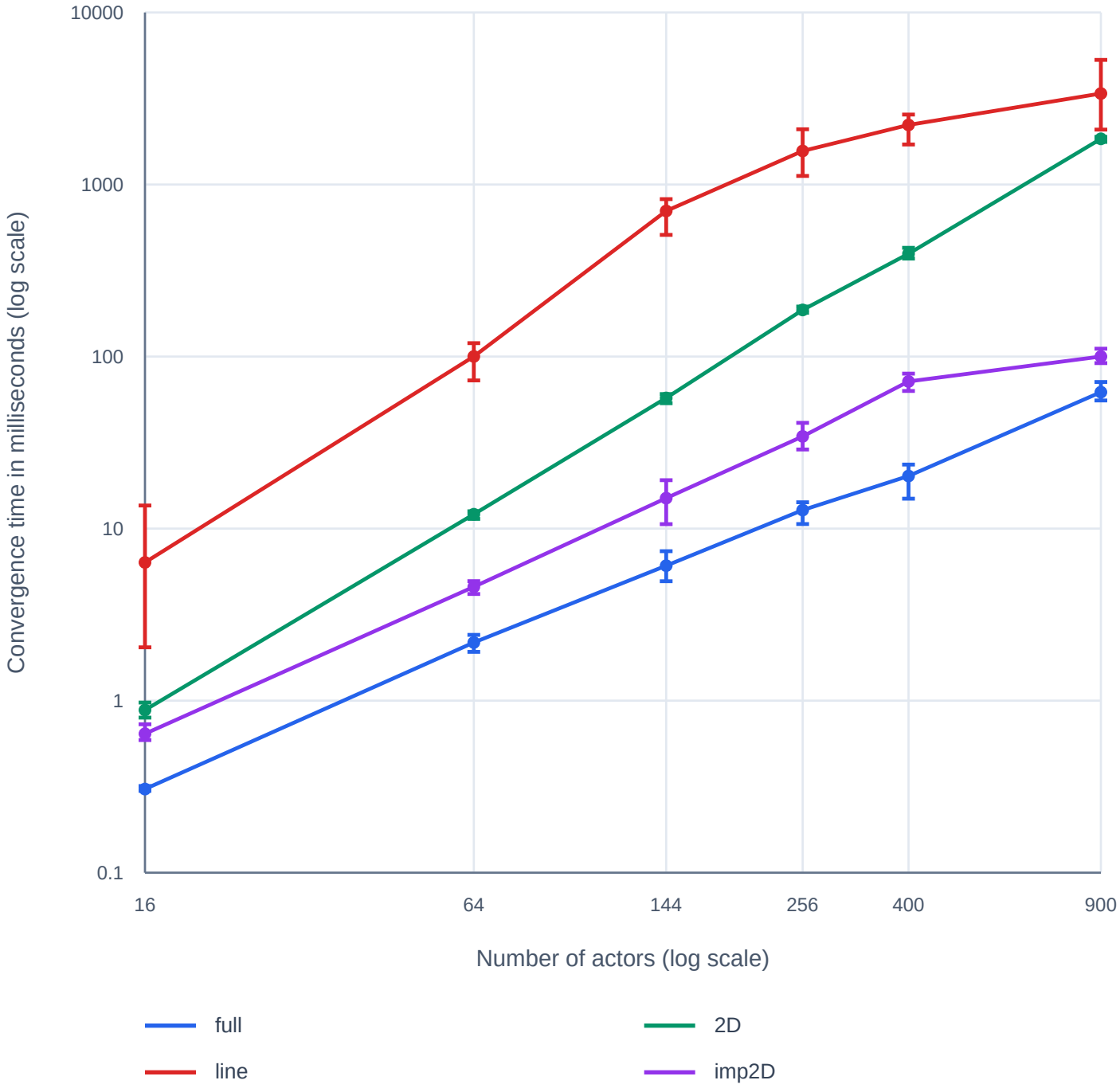
Full rumor coverage vs. actor count | logarithmic axes



Dots: mean of all three successful trials. Bars: observed min/max, not confidence intervals. An X indicates at least one timeout; the point is drawn at the time limit and is excluded from the curve.

Push-sum local stability

First local termination vs. actor count | logarithmic axes



Dots: mean of all three successful trials. Bars: observed min/max, not confidence intervals. An X indicates at least one timeout; the point is drawn at the time limit and is excluded from the curve.

Results and interpretation

Success rates, tested network sizes and observed limitations

Largest measured successful sizes

Topology	Gossip	Push-sum	Success criterion
full	900	900	3 / 3 runs
2D	900	900	3 / 3 runs
line	900	900	3 / 3 runs
imp2D	900	900	3 / 3 runs

These are the largest tested sizes that converged in every trial, not proven capacity limits. The largest attempted size is 900 actors for every combination.

Observed findings

At 900 actors, gossip: full: 3/3 successes, mean 18.67 ms; 2D: 3/3 successes, mean 45.66 ms; line: 3/3 successes, mean 1199.12 ms; imp2D: 3/3 successes, mean 23.69 ms.

At 900 actors, push-sum: full: 3/3 successes, mean 62.10 ms; 2D: 3/3 successes, mean 1843.40 ms; line: 3/3 successes, mean 3379.83 ms; imp2D: 3/3 successes, mean 100.05 ms.

Long paths restrict rumor propagation and slow ratio mixing. The full graph and random shortcut edges generally spread information faster. Local push-sum stability can also occur early, before accurate global mixing; a fast run with a large ratio error is not successful averaging. The imperfect grid has one fixed directed extra edge per actor, selected uniformly from non-self, non-grid neighbors; the added edge is not automatically reciprocal.

The two algorithm curves are not directly comparable performance rankings: gossip measures coverage with timer-driven sends, while push-sum measures first local numerical stability using receive-driven sends. Differences reflect both protocol work and the termination criterion.

Numerical accuracy and limits

The correct average is (N+1)/2. Across locally converged push-sum trials, the largest terminating actor estimate error was 350. This diagnostic describes the first terminating actor only; it is not a network-wide error bound. Some terminating actors have large errors, exposing the weakness of local stability as a global stopping rule. The required local change threshold is not a global error bound.

Timeouts are recorded as failures and excluded from means. Global mode is available for stronger experiments, but it changes local termination semantics and should be labeled separately. Three trials on one machine provide an initial comparison; larger samples, additional sizes and machines would strengthen conclusions. Raw data and scripts are included so results can be reproduced and extended.